

Jens Laufer

FDE Lead — architecture, hands-on delivery and LLM agents in production, owned end to end with the client

Architecture & technical discovery · LLM agents, tool use & MCP · evaluation, quality gates & drift control · Python · Java / Spring Boot · Vue / TypeScript · REST & enterprise integration (SAP, ESB, canonical data models) · Docker / Kubernetes · CI/CD · observability · OAuth2 / OIDC

I own a technical engagement from the first conversation to the handover: discovery, target-state architecture, the trade-offs, and the code. Since 10/2025 I build and run LLM agent systems in production — an autonomous AI business partner that plans, writes code, tests and deploys around the clock, Fabrik, a harness where worker agents ship software in parallel, and an agent that runs the finance back office of a GmbH. The hard part there is control, not prompting: quality gates an agent must pass before it may ship, approval gates before anything irreversible, observability over every run, and checks that catch an agent drifting from the spec. Before that, 29 years of back-end, full-stack and integration work inside client teams — Deutsche Telekom, Commerzbank, Volkswagen, 1&1, DVAG, Brunata. Freelance since 1997, so scoping, pushing back on what the evidence does not support, delivery and production support have always been one conversation.

AVAILABLE immediately	WORKS remote only	REMOTE SHARE 100 % 0 % on-site	BASED IN Karlstein am Main Germany
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FOCUS

End-to-end technical ownership: freelance since 1997, embedded in the client's team — technical discovery, architecture and design decisions, delivery, production support and handover, on site or remote.

AI-assisted engineering with controls, in production since 10/2025: agent loop with stop conditions, tool use over MCP, layered memory, RAG, quality gates and approval gates, observability over every run, budget and rate-limit control with model routing, and recovery after a killed run.

Verification against agent drift: agents' output is checked against the spec and the diff, not against their own report — a test must reproduce a bug red before the fix may turn it green.

Agent harnesses in daily production use: Fabrik (parallel worker agents, ticket to PR), an unattended coding pipeline (759 tasks to date) and an LLM agent that runs the finance back office of a GmbH with a human approving every filing.

Enterprise integration: invoice handover to **SAP** over an ESB (1&1), master data migration on a canonical XML model (Commerzbank), an industry file format in heating cost billing (Brunata).

Prototype to production: took over a running prototype at Deutsche Telekom and put it live — Docker/Kubernetes, tracing and CI included.

ROLES

- Full-stack engineering
- Solution design
- Backend engineering
- DevOps
- Build & configuration management

Data analysis

Prototyping

Machine learning

Jenkins plugin development

Maven plugin development

SKILLS

DEVOPS / CLOUD

Docker

Kubernetes

Rancher

GitLab CI

Jenkins

Maven

Grunt

npm

Artifactory

AWS (EC2, S3, SageMaker, VPC)

Microsoft Azure

DigitalOcean

MICROSERVICES / BACKEND

Spring Boot

Spring

Spring Data

Spring Integration

Spring Security

OAuth2

Microservices

Feign

Sleuth

Zipkin

Python Eve

Python Flask

Java EE

EJB

FRONTEND

Vue.js

React

AngularJS

GWT

Quasar

HTML

CSS

Bootstrap

SECURITY / IAM

Keycloak

Spring Security

OAuth2

MACHINE LEARNING

scikit-learn

Keras

DeepLearning4J

CNN

Transfer Learning

Supervised/Unsupervised

k-means

PCA

t-SNE

Reinforcement Learning

OpenAI Gym

Topic Modeling

TOOLING / OTHER

Git

GitLab

Subversion

Jira

Confluence

Kibana

ANTLR

JavaCC

ZXing (QR)

PayPal API

RDF/XML/JSON

SAP (BAPI)

Phonegap

JBoss

Tomcat

LANGUAGES

Java

JavaScript

TypeScript

Python

R

Groovy

API / INTEGRATION

REST

Swagger

OpenAPI

SOAP

Webservices

CXF

XFire

Kafka

ServiceMix

PERSISTENCE / DATABASES

MySQL

MongoDB

PostgreSQL

Oracle

Neo4j

JPA

Hibernate

TESTING

JUnit

TDD

Mockito

Selenium

DBUnit

Playwright

DATA SCIENCE

pandas

numpy

dplyr

Jupyter

RMarkdown

ggplot

seaborn

Vega-Lite

A/B Testing

Webscraping

PROJECT HISTORY (SELECTION FROM 2009)

since 04/2026
ongoing

Fabrik — agent harness for parallel software delivery

Fabrik (fabrikhq.com) · Solytics GmbH · Remote — Software / AI
Harness engineering · Architecture · Full-stack engineering

A harness in which worker agents turn tickets into pull requests in parallel (fabrikhq.com). A **control plane** (FastAPI, Vue) spawns one agent container per repository through the Docker socket. Each agent runs a fixed loop — ticket sync (GitHub, GitLab, Jira), an actionability review that rejects vague tickets, a plan, test-first implementation, a **quality gate** in a clean sidecar container, then the PR; it never merges on its own. Energy budget per agent, feedback from PR comments back into the loop, one VPS per customer behind Caddy.

Claude / Anthropic API Multi-Agent-Orchestration Python FastAPI Vue 3 Docker Caddy GitHub Actions
SQLite TDD

since 03/2026
ongoing

Autonomous coding pipeline — agents work a task queue unattended

agent-tasks · Solytics GmbH · Remote — Software / AI
Harness engineering · Agent engineering · DevOps

Queue-driven harness in which coding agents work through tasks with nobody at the keyboard. A task file names repository, scope, acceptance criteria and model; a runner on own hardware executes it in an isolated worktree, runs the **quality gate** (unit tests, build, review) before any push and opens a pull request — merging stays with a human. Model chosen per task against a 5-hour token budget, hard timeouts per run, at most one new pull request per repository per night. **759 tasks** written to date across the own product repositories.

Claude / Anthropic API Python Docker TDD systemd

since 10/2025
ongoing

Autonomous AI business partner — Harness & agentic engineering

Solytics GmbH · Remote — AI / Automation
Harness engineering · Agent engineering · AI system architecture · DevOps

A self-directing agent built on Claude that plans, writes code, tests, deploys and reports around the clock. Built as a **harness engineer** — the engineering around the model: the **agent loop** and its stop conditions, **tool use over MCP**, layered **memory** (durable facts, running context, session state), **approval gates** for anything irreversible, **observability** over every run, **budget and rate-limit control** with model routing, plus recovery when a run is killed mid-flight. More than 40 purpose-built checks catch the agent's mistakes before they reach a human. Event-driven over Telegram, GitHub and systemd; runs on own hardware behind a local multi-model proxy.

Claude / Anthropic API LLM-Agenten Agent Harness Tool Use / MCP RAG Prompt Engineering Python
systemd Telegram Bot API Docker TDD

since 03/2026
ongoing

Verification and drift controls for LLM agents

Solytics GmbH · Remote — AI / Automation
Evaluation & verification · Reliability engineering

More than 40 purpose-built checks that test an agent's work against the world instead of against its own report: every link in an outgoing message is resolved before it is sent, promises the agent made are tracked until delivered, messages without an answer are flagged, pull-request mergeability is computed with git instead of read from a cached API value, and a test run that never started fails instead of passing silently. Each check fails closed — unknown is never reported as healthy — and has its own tests. Watchdogs over the agent's services and credentials, with a second alert path for when the primary channel is the one that broke.

Claude / Anthropic API Python Docker TDD

since 03/2026
ongoing

LLM agent in the finance back office of a GmbH

Solytics GmbH · Remote — Finance / Compliance
Agent integration · Domain design · Back-end engineering

An agent that runs the bookkeeping of a German GmbH end to end, with a human approving every filing. It files bank statements and receipts, books double-entry on SKR04 through a purpose-built CLI, reconciles every account against the statement, and checks VAT and payroll returns against the journal before they are submitted — once catching a field that would have lost the input tax. Tools beneath it: an API for **XRechnung 3.0** and **ZUGFeRD 2.3** with five validation layers (Pydantic, BR-DE rules, arithmetic, XSD, Schematron), balance sheet, P&L, E-Bilanz and **DATEV export**.

Python Pydantic XRechnung / ZUGFeRD (EN 16931) Polars FastAPI Docker TDD

since 03/2026
ongoing

Launch Kit — multi-tenant identity and email platform

Launch Kit · Solytics GmbH · Remote — SaaS / IAM
Architecture · Back-end engineering · Security

OIDC-compliant auth server with email marketing as one Go binary and a Vue admin UI — a drop-in for Keycloak or Auth0. **OAuth2 with PKCE**, RS256 JWTs with a per-tenant RSA key, Login with Google, role-based access. Contacts, segments and campaigns with async sending, unsubscribe handling and GDPR consent. Strict tenant isolation; apps validate tokens over JWKS only and swap the provider by changing an environment variable. Runs the customer funnel of several own products.

Go OpenID Connect OAuth2 JWT Vue 3 SQLite Docker Caddy Playwright

07/2024–06/2025
12 months

Digital property profile for owners

verkaufe.immobilien · Remote — Internet
Full-stack engineering · Solution design · DevOps

Full-stack development of a digital property profile that owners create and maintain themselves: Vue.js front-end, Python back-end (Eve) with a REST API and MongoDB. Built and maintained the GitLab CI pipelines for automated build and deployment. Worked to DevOps principles, owning build and operation of the services in Docker. Test-driven development, automated end-to-end tests with Playwright.

Python Python Eve MongoDB TDD Vue.js Docker GitLab GitLab CI REST Playwright

07/2022–04/2024
22 months

Front- and back-end, heating cost billing

Brunata · Remote — Industry
Full-stack engineering · Solution design · DevOps

Production server-side services and APIs in Java and Spring Boot: a microservice that parses and generates heating-cost files in the Haweiko-Arge industry format. Designed and built in a microservices architecture with REST interfaces (OpenAPI, Swagger), secured with Spring Security and Keycloak, persisted in MongoDB. Built and maintained the GitLab CI pipelines for automated build and deployment. Worked to DevOps principles, owning build and operation of the services in Docker. Test-driven development (JUnit, Mockito), automated end-to-end tests with Playwright. Front-end development.

Java Spring Boot MongoDB JUnit TDD React Docker GitLab GitLab CI Spring Security
Keycloak Mockito OpenAPI Swagger REST Microservices Playwright

01/2022–06/2022

6 months

Payment back-end migration Java EE → Spring Boot

Condor · Remote — Marketing

Backend engineering

Migrated an EJB-based Java EE payment back-end to Spring Boot and shipped it as a Docker container.

Java Java EE JPA Spring Boot MySQL Docker Mockito JUnit TDD

12/2019–12/2020

13 months

Web-based configuration of telecommunications devices

Deutsche Telekom · Darmstadt/Remote — Telecommunications

DevOps · Build & configuration management · Full-stack engineering

Took over an existing prototype for configuring telecommunications devices over a web interface, developed it further and moved it into production. Production server-side services and APIs in Java and Spring Boot in a microservices architecture (REST, Feign, Spring Data/JPA), Vue.js front-end, secured with Keycloak, persisted in MySQL and MongoDB. Worked to DevOps principles, owning build, operation and monitoring of the services: Docker and Kubernetes (Rancher), logging and tracing via Kibana, Sleuth and Zipkin; built and maintained the GitLab CI pipelines.

Java Spring Boot Docker Kubernetes Microservices REST Spring Data JPA JUnit TDD
Mockito Vue.js MySQL MongoDB HTML CSS GitLab GitLab CI Rancher Kibana Feign
Keycloak Sleuth Zipkin

08/2017–03/2019

20 months

Connected Van — driver's logbook

Volkswagen · Remote — Automotive

Backend engineering

Microservices for vehicle geolocation and a fuel logbook.

Spring Boot Docker Kubernetes Microservices REST Java Maven Swagger OpenAPI Feign
Sleuth Zipkin MySQL MongoDB Jira Confluence JUnit TDD Mockito Microsoft Azure
Kibana Spring Git Kafka

07/2015–07/2017

25 months

Invoice management platform (SME)

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

Platform covering the full invoice life cycle (creation, payment, dunning, debt collection); a range of Spring Boot microservices.

Spring Boot Docker Kubernetes Microservices REST Java Maven Spring Security OAuth2
Spring Integration MongoDB Jira Confluence JUnit TDD Mockito Selenium AngularJS Grunt
npm Spring Git

04/2010–09/2010

6 months

Web services on an ESB (mediation and rating)

1&1 · Montabaur — Telecommunications

Backend engineering

Software design, web services, maintenance of the rating engine; web services that hand invoices over to **SAP**.

Java Spring Spring Security JPA Hibernate Maven JUnit TDD Mockito DBUnit ServiceMix
Webservices CXF XFire SAP-Anbindung SAP BAPI JBoss Oracle Subversion (SVN)

05/2009–03/2010

11 months

Master data migration at a major bank

Commerzbank · Frankfurt a. Main — Banking, Finance

Backend engineering

Workflows for persisting master data, a canonical XML model, RDF description for machine processing.

- Java
- Java EE
- IBM MDM (WCC)
- EJB
- JPA
- RDF
- XML
- Oracle
- Subversion (SVN)

CONTACT & CREDENTIALS

CONTACT & PROFILES

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PUBLICATION	Towards Data Science — author

EDUCATION

11/1997	Dipl.-Ing. (Univ.) Electrical Engineering (MSc equivalent) University of Erlangen-Nuremberg
05/1989	Abitur (university entrance qualification) Gymnasium am Romäusring, Villingen-Schwenningen

CERTIFICATES

12/2018	Machine Learning Engineer Nanodegree Udacity Verify →
08/2017	Data Analyst Nanodegree Udacity Verify →